

Models Documentation (All Apps)

This document contains model-level documentation for all 'models.py' files in the repository, including fields, relationships, constraints, and helper methods.

Files Covered

- 'Employee/models.py': 2 model(s)
- 'Department/models.py': 1 model(s)
- 'Role/models.py': 1 model(s)
- 'Project/models.py': 1 model(s)
- 'Leave/models.py': 1 model(s)
- 'Attendance/models.py': 1 model(s)
- 'Dashboard/models.py': 0 model(s)
- 'Login_App/models.py': 0 model(s)

Model Relationship Diagram (ER)

```
'''mermaid
erDiagram
    BUSINESSPROFILE ||--o{ EMPLOYEE : owns
    BUSINESSPROFILE ||--o{ DEPARTMENT : has
    BUSINESSPROFILE ||--o{ JOB : has
    BUSINESSPROFILE ||--o{ PROJECT : has
    BUSINESSPROFILE ||--o{ LEAVE : has
    BUSINESSPROFILE ||--o{ ATTENDANCE : has
    DEPARTMENT ||--o{ EMPLOYEE : contains
    JOB ||--o{ EMPLOYEE : assigns
    EMPLOYEE ||--o{ EMPLOYEE : manages
    EMPLOYEE ||--o{ LEAVE : requests
    EMPLOYEE ||--o{ LEAVE : approves
    EMPLOYEE ||--o{ ATTENDANCE : marks
    PROJECT }o--o{ EMPLOYEE : assigned
'''
```

File: 'Employee/models.py'

Model: 'BusinessProfile'

- **Base classes**: 'models.Model'
- **Source location**: 'Employee/models.py:7-14'

Fields

Field	Type	Key options
'business_name'	'CharField'	max_length=100
'business_address'	'CharField'	max_length=255
'business_phone'	'CharField'	max_length=15
'business_email'	'EmailField'	-

Meta options

No Meta options defined.

Methods

- '__str__()' ('Employee/models.py:13-14')

Source snippet

```
'''python
class BusinessProfile(models.Model):
    business_name = models.CharField(max_length=100)
    business_address = models.CharField(max_length=255)
    business_phone = models.CharField(max_length=15)
    business_email = models.EmailField()
```

```

def __str__(self):
    return self.business_name
'''

### Model: 'Employee'
- **Base classes**: 'AbstractUser'
- **Source location**: 'Employee/models.py:16-67'

#### Fields
| Field | Type | Key options |
|---|---|---|
| 'role' | 'CharField' | max_length=10; default='employee'; choices=ROLE_CHOICES |
| 'business_profile' | 'ForeignKey' | null=True; blank=True; on_delete=models.CASCADE;
related_name='employees'; args=BusinessProfile |
| 'phone' | 'CharField' | max_length=15; blank=True |
| 'hire_date' | 'DateField' | default=now |
| 'salary' | 'DecimalField' | null=True; blank=True |
| 'department' | 'ForeignKey' | null=True; blank=True; on_delete=models.SET_NULL;
related_name='employees'; args=Department |
| 'job' | 'ForeignKey' | null=True; blank=True; on_delete=models.SET_NULL;
related_name='employees'; args=Job |
| 'manager' | 'ForeignKey' | null=True; blank=True; on_delete=models.SET_NULL;
related_name='team_members'; args='self' |

#### Meta options
_No Meta options defined._

#### Methods
- '__str__()' ('Employee/models.py:66-67')

#### Source snippet
'''python
class Employee(AbstractUser):
    ROLE_CHOICES = (
        ('employee', 'Employee'),
        ('owner', 'Owner'),
    )

    role = models.CharField(max_length=10, choices=ROLE_CHOICES, default='employee')

    business_profile = models.ForeignKey(
        BusinessProfile,
        null= True,
        blank= True,
        on_delete=models.CASCADE,
        related_name='employees'
    )

    phone = models.CharField(max_length=15, blank=True)
    hire_date = models.DateField(default=now)

    salary = models.DecimalField(
        max_digits=10,
        decimal_places=2,
        null=True,
        blank=True
    )

    department = models.ForeignKey(
        Department,
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='employees'

```

```

)

job = models.ForeignKey(
    Job,
    on_delete=models.SET_NULL,
    null=True,
    blank=True,
    related_name='employees'
)

manager = models.ForeignKey(
    'self',
    on_delete=models.SET_NULL,
    null=True,
    blank=True,
    related_name='team_members'
)

def __str__(self):
    return self.username
'''

---

## File: 'Department/models.py'

### Model: 'Department'
- **Base classes**: 'models.Model'
- **Source location**: 'Department/models.py:4-17'

#### Fields
| Field | Type | Key options |
|---|---|---|
| 'name' | 'CharField' | max_length=100 |
| 'description' | 'TextField' | blank=True |
| 'business_profile' | 'ForeignKey' | on_delete=models.CASCADE; related_name='departments'; args='Employee.BusinessProfile' |

#### Meta options
_No Meta options defined._

#### Methods
- 'total_employees()' ('Department/models.py:13-14')
- '__str__()' ('Department/models.py:16-17')

#### Source snippet
'''python
class Department(models.Model):
    name = models.CharField(max_length=100)
    description = models.TextField(blank=True)

    business_profile = models.ForeignKey(
        'Employee.BusinessProfile',
        on_delete=models.CASCADE,
        related_name='departments'
    )
    def total_employees(self):
        return self.employees.count()

    def __str__(self):
        return self.name
'''

---

```

```

## File: 'Role/models.py'

### Model: 'Job'
- **Base classes**: 'models.Model'
- **Source location**: 'Role/models.py:2-11'

#### Fields
| Field | Type | Key options |
|---|---|---|
| 'title' | 'CharField' | max_length=100 |
| 'description' | 'TextField' | blank=True |
| 'business_profile' | 'ForeignKey' | on_delete=models.CASCADE; related_name='jobs';
args='Employee.BusinessProfile' |

#### Meta options
_No Meta options defined._

#### Methods
- '__str__()' ('Role/models.py:10-11')

#### Source snippet
'''python
class Job(models.Model):
    title = models.CharField(max_length=100)
    description = models.TextField(blank=True)
    business_profile = models.ForeignKey(
        'Employee.BusinessProfile',
        on_delete=models.CASCADE,
        related_name='jobs'
    )
    def __str__(self):
        return self.title
'''

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## File: 'Project/models.py'

### Model: 'Project'
- **Base classes**: 'models.Model'
- **Source location**: 'Project/models.py:3-23'

#### Fields
| Field | Type | Key options |
|---|---|---|
| 'name' | 'CharField' | max_length=100 |
| 'description' | 'TextField' | blank=True |
| 'business_profile' | 'ForeignKey' | on_delete=models.CASCADE; related_name='projects';
args='Employee.BusinessProfile' |
| 'employees' | 'ManyToManyField' | blank=True; related_name='projects'; args=Employee |
| 'start_date' | 'DateField' | - |
| 'end_date' | 'DateField' | null=True; blank=True |

#### Meta options
_No Meta options defined._

#### Methods
- '__str__()' ('Project/models.py:22-23')

#### Source snippet
'''python
class Project(models.Model):
    name = models.CharField(max_length=100)

```

```

description = models.TextField(blank=True)

business_profile = models.ForeignKey(
    'Employee.BusinessProfile',
    on_delete=models.CASCADE,
    related_name='projects'
)

employees = models.ManyToManyField(
    Employee,
    related_name='projects',
    blank=True
)

start_date = models.DateField()
end_date = models.DateField(null=True, blank=True)

def __str__(self):
    return self.name
'''
'''

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## File: 'Leave/models.py'

### Model: 'Leave'
- **Base classes**: 'models.Model'
- **Source location**: 'Leave/models.py:4-65'

#### Fields
| Field | Type | Key options |
|---|---|---|
| 'employee' | 'ForeignKey' | on_delete=models.CASCADE; related_name='leaves'; args=Employee |
| 'business_profile' | 'ForeignKey' | on_delete=models.CASCADE; related_name='leaves'; args='Employee.BusinessProfile' |
| 'leave_type' | 'CharField' | max_length=10; choices=LEAVE_TYPE |
| 'start_date' | 'DateField' | - |
| 'end_date' | 'DateField' | - |
| 'reason' | 'TextField' | - |
| 'status' | 'CharField' | max_length=10; default='pending'; choices=LEAVE_STATUS |
| 'approved_by' | 'ForeignKey' | null=True; blank=True; on_delete=models.SET_NULL; related_name='approved_leaves'; args='Employee.Employee' |
| 'applied_on' | 'DateTimeField' | - |

#### Meta options
_No Meta options defined._

#### Methods
- 'clean()' ('Leave/models.py:55-62')
- '__str__()' ('Leave/models.py:64-65')

#### Source snippet
'''python
class Leave(models.Model):
    LEAVE_STATUS = (
        ('pending', 'Pending'),
        ('approved', 'Approved'),
        ('rejected', 'Rejected'),
    )

    LEAVE_TYPE = (
        ('casual', 'Casual'),
        ('sick', 'Sick'),
        ('paid', 'Paid'),

```

```

        ('unpaid', 'Unpaid'),
    )

    employee = models.ForeignKey(
        Employee,
        on_delete=models.CASCADE,
        related_name='leaves'
    )

    business_profile = models.ForeignKey(
        'Employee.BusinessProfile',
        on_delete=models.CASCADE,
        related_name='leaves'
    )

    leave_type = models.CharField(
        max_length=10,
        choices=LEAVE_TYPE
    )

    start_date = models.DateField()
    end_date = models.DateField()
    reason = models.TextField()

    status = models.CharField(
        max_length=10,
        choices=LEAVE_STATUS,
        default='pending'
    )

    approved_by = models.ForeignKey(
        'Employee.Employee',
        on_delete=models.SET_NULL,
        null=True,
        blank=True,
        related_name='approved_leaves'
    )

    applied_on = models.DateTimeField(auto_now_add=True)

    def clean(self):
        if self.end_date < self.start_date:
            raise ValidationError("End date cannot be before start date")

        if self.employee.business_profile != self.business_profile:
            raise ValidationError(
                "Employee must belong to the same business as the leave"
            )

    def __str__(self):
        return f"{self.employee.username} ({self.leave_type}) - {self.start_date} to {self.end_date}"
'''
'''

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## File: 'Attendance/models.py'

### Model: 'Attendance'
- **Base classes**: 'models.Model'
- **Source location**: 'Attendance/models.py:5-62'

#### Fields
| Field | Type | Key options |

```

```
|---|---|---|
| 'employee' | 'ForeignKey' | on_delete=models.CASCADE; related_name='attendance';
args='Employee.Employee' |
| 'business_profile' | 'ForeignKey' | null=True; blank=True; on_delete=models.CASCADE;
related_name='attendance'; args='Employee.BusinessProfile' |
| 'date' | 'DateField' | - |
| 'check_in' | 'TimeField' | null=True; blank=True |
| 'check_out' | 'TimeField' | null=True; blank=True |
| 'status' | 'CharField' | max_length=10; default='present'; choices=STATUS_CHOICES |
```

Meta options

```
- 'unique_together' = (('employee', 'date'))
- 'ordering' = ''
```

Methods

```
- 'clean()' ('Attendance/models.py:40-49')
- 'save()' ('Attendance/models.py:51-54')
- 'Total_hours()' ('Attendance/models.py:56-60')
- '__str__()' ('Attendance/models.py:61-62')
```

Source snippet

```
'''python
class Attendance(models.Model):
    STATUS_CHOICES = (
        ('present', 'Present'),
        ('absent', 'Absent'),
        ('half_day', 'Half Day'),
    )

    employee = models.ForeignKey(
        'Employee.Employee',
        on_delete=models.CASCADE,
        related_name='attendance'
    )

    business_profile = models.ForeignKey(
        'Employee.BusinessProfile',
        on_delete=models.CASCADE,
        related_name='attendance',
        null=True,
        blank=True
    )

    date = models.DateField()
    check_in = models.TimeField(null=True, blank=True)
    check_out = models.TimeField(null=True, blank=True)

    status = models.CharField(
        max_length=10,
        choices=STATUS_CHOICES,
        default='present'
    )

    class Meta:
        unique_together = ('employee', 'date')
        ordering = ['-date']

    def clean(self):
        if self.check_in and self.check_out:
            if self.check_out < self.check_in:
                raise ValidationError("Check-out time cannot be before check-in")

        if self.employee_id and self.business_profile_id:
            if self.employee.business_profile_id != self.business_profile_id:
```

```

        raise ValidationError(
            "Employee must belong to the same business as attendance record"
        )

    def save(self, *args, **kwargs):
        if not self.business_profile and self.employee:
            self.business_profile = self.employee.business_profile
        super().save(*args, **kwargs)

    def Total_hours(self):
        if self.check_in and self.check_out:
            delta = datetime.combine(date.min, self.check_out) - datetime.combine(date.min,
self.check_in)
            return delta
        return None
    def __str__(self):
        return f"{self.employee.username} - {self.date}"
'''

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## File: 'Dashboard/models.py'

_No model classes defined in this file._

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## File: 'Login_App/models.py'

_No model classes defined in this file._

```